

# Topology

## Theatre nodes (×12)

Each theatre is its own subnet, NAT'd by its own GL-iNet **A-1300 (Slate Plus)** router. The router runs Tailscale as a subnet router (LAN + WAN side routing enabled), advertising the theatre's /24 .

**Physical wiring:** the A-1300 has 1 WAN + 2 LAN gigabit ports. Both LAN ports are used:

- **LAN 1 → ATEM Mini Extreme ISO directly.** Dedicated port, so the router's other LAN traffic (rclone syncs, control laptop, etc.) never shares a switch segment with it. The ATEM carries the heaviest sustained traffic of any theatre device — the near-real-time ISO ingest pull (see docs/atem-iso-ingest.md ) is a continuous 50-90 Mbps flow, well above BirdDog Play's — so it gets the dedicated port.
- **LAN 2 → 8-port unmanaged Netgear switch,** which fans out to everything else: BirdDog Play, PowerPoint Main/Backup, VT Main/Backup, Control laptop (5 devices, 3 spare switch ports).

All devices stay on the same 192.168.X.0/24 — this is a physical port split for traffic separation, not a VLAN/subnet split.

| Role                                | Address      | Physical connection           |
|-------------------------------------|--------------|-------------------------------|
| GL-iNet A-1300 router (LAN gateway) | 192.168.X.1  | —                             |
| ATEM Mini Extreme ISO               | 192.168.X.2  | A-1300 LAN 1 (direct)         |
| BirdDog Play                        | 192.168.X.20 | Netgear switch → A-1300 LAN 2 |
| PowerPoint Main                     | 192.168.X.5  | Netgear switch → A-1300 LAN 2 |
| PowerPoint Backup                   | 192.168.X.6  | Netgear switch → A-1300 LAN 2 |
| VT Main                             | 192.168.X.7  | Netgear switch → A-1300 LAN 2 |
| VT Backup                           | 192.168.X.8  | Netgear switch → A-1300 LAN 2 |
| Control laptop                      | 192.168.X.10 | Netgear switch → A-1300 LAN 2 |

X per the subnet map in the top-level README: Theatre 1 → 2 , Theatre 2 → 3 , ... Theatre 12 → 13 .

The ATEM's built-in FTP server (its ISO recording drive) is pulled continuously by the mothership over Tailscale subnet routing — see docs/atem-iso-ingest.md .

**Alternative to the GL-iNet router,** documented but not recommended: plain switch + Tailscale on generic Linux — see docs/topology-alternative-tailscale-switches.md .

## Uplink VLAN grouping

Theatres are grouped 3-per-physical-VLAN purely to organize uplink cabling back to the mothership — **this grouping does not itself provide cross-theatre isolation.** Each GL-iNet A-1300 already NATs its own

theatre LAN, and once every router joins the same tailnet, routes are reachable tailnet-wide unless Tailscale ACLs restrict them (see `config/tailscale-acl.json` ).

**These VLANs are venue-supplied, not our infrastructure** — each capped at 1 Gbps, each port expensive. The 4-VLAN/3-theatres-each split below is safe but conservative; the real bandwidth math supports consolidating to 2 VLANs (6 theatres each) with genuine margin to spare — see `docs/bandwidth-analysis.md` for the full model and recommendation.

| VLAN | Theatres   | Subnets             |
|------|------------|---------------------|
| 1    | 1, 2, 3    | .2.x, .3.x, .4.x    |
| 2    | 4, 5, 6    | .5.x, .6.x, .7.x    |
| 3    | 7, 8, 9    | .8.x, .9.x, .10.x   |
| 4    | 10, 11, 12 | .11.x, .12.x, .13.x |

## Mothership — 192.168.1.x

Ubiquiti **Cloud Gateway**; all 4 VLANs terminate here via separate links. Also hosts the 4x "ready room" computers (physical, untouched) and the hardwired WAN.

**Consolidated services server (Unraid)**. Everything else that used to be described as separate mothership boxes — Nextcloud, Restreamer, BirdDog Central, the NDI Discovery Server, a self-hosted DERP server, the ATEM ISO ingest pipeline, the VMix record ingest pipeline, a self-hosted UniFi Controller, self-hosted GLKVM-Cloud, ATEM Overseer + ATEM Fleet Admin (fleet monitoring and provisioning for the 12 theatre ATEMs), and now Flock (BirdDog Play fleet manager) — runs on a single physical server instead, virtualized with Unraid. (An earlier revision also carried a mothership-local VMix instance VM; it was removed from the design — the theatre program feeds originate from the VMix *node* PCs, and no mothership role for a local VMix instance was ever established, see `docs/open-questions.md` #10.) The server already exists (no procurement needed) — known so far: **2x gigabit NICs (bonded — see below), a "decent" discrete GPU** (on hand, though nothing in the design requires it any more — see the GPU section of `docs/server-specification.md` ). Full spec (CPU/motherboard, RAM) is TBD — see `docs/open-questions.md` and the calculated target minimum spec (storage pool layout, RAM, CPU, GPU tier, all worked from this box's actual workload) in `docs/server-specification.md` .

| Workload             | Runs as          | Notes                                                                                                                                                                                                  |
|----------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BirdDog Central      | Windows VM       | Windows-only (tech specs); the design's only VM                                                                                                                                                        |
| Nextcloud            | Docker container | official image + MariaDB + Redis (see <code>config/docker-compose.yml</code> )                                                                                                                         |
| Restreamer           | Docker container | official <code>datarhei/restreamer</code> image                                                                                                                                                        |
| NDI Discovery Server | Docker container | see <code>docs/streaming-flow.md</code>                                                                                                                                                                |
| DERP server          | Docker container | self-hosted relay fallback — see <code>docs/tailscale.md</code>                                                                                                                                        |
| ATEM ISO Ingest      | Docker container | pulls all 12 theatres' ATEM ISO recordings via FTP, dual-writing to Nextcloud and the edit-suite NAS in the same pass — see <code>docs/atem-iso-ingest.md</code> and <code>docs/live-editing.md</code> |
| VMix Record Ingest   | Docker container | pulls all 4 VMix PCs' recordings via SMB, same dual-write pattern — see <code>docs/vmix-record-ingest.md</code>                                                                                        |
| UniFi Controller     | Docker container | self-hosted UniFi Network Application — local management/backup console for the Cloud Gateway, independent of cloud.ui.com — see <code>config/unifi/README.md</code>                                   |

| Workload                          | Runs as                           | Notes                                                                                                                                                                                     |
|-----------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GLKVM-Cloud<br>( rttys + coturn ) | 2 Docker containers               | self-hosted remote administration (web UI + SSH terminal) for the 14 GL-iNet routers — see docs/glkvcloud.md                                                                              |
| ATEM Overseer                     | Docker container                  | fleet monitoring/tally dashboard for all 12 theatres' ATEMs — receives each theatre's monitoring stream, see docs/bandwidth-analysis.md                                                   |
| ATEM Fleet Admin                  | Docker container                  | bulk ATEM provisioning — model-aware config forms, live network apply via atem-connection over the same Tailscale subnet routing the ISO ingest pipeline uses, or XML+media folder export |
| Flock                             | Docker container                  | BirdDog Play fleet manager — LAN discovery, tag-based grouping, BirdUI-parity settings, batch edits; receives each theatre's BirdDog Play preview stream, see docs/bandwidth-analysis.md  |
| Tailscale subnet router           | Docker container, host networking | advertises 192.168.1.0/24 ; replaces the earlier "Nextcloud host" plan now that Nextcloud itself is a container on this same box                                                          |

**Bonded NICs.** Decided: bond the 2 gigabit NICs (802.3ad/LACP) rather than split traffic across them — SRT bandwidth isn't constant, and LACP's hash-based distribution spreads the many *simultaneous* flows this box handles (12 theatres' SRT fan-out, each to a different destination, the 12 incoming Overseer monitoring streams and Flock SRT previews, plus rclone/Nextcloud/BirdDog Central/NDI Discovery Server/DERP traffic) across both links by destination-IP/port hash — giving real aggregate headroom for bursty traffic, not just failover. Note this does **not** speed up any single flow; the benefit comes from having many distinct flows to hash across.

**Assuming the Cloud Gateway supports LAG** on the ports the Unraid server lands on (only UDM Pro/SE/Pro Max, UXG Enterprise, and EFG support port aggregation — not base Cloud Gateway models). If it turns out this one doesn't: drop an intermediate LACP-capable switch in between and bond through that instead — the bond doesn't care which device terminates it, only that something in the path speaks LACP, so this is a cheap fallback rather than a blocker.

One thing that does need doing regardless of which device terminates the bond: **LACP rate/hash policy must match on both ends.** Ubiquiti gear hardcodes LACP rate `fast` and hash policy `layer3+4` ; Unraid's bonding defaults differ ( `slow` rate, layer2 hash) and need to be set to match, or the bond won't form correctly.

**Why Unraid over TrueNAS SCALE:** both support Docker and Windows VMs, and Unraid has the more polished one-click container experience (Community Applications) — useful since this may be maintained on-site by AV staff rather than a Linux admin. Trade-off: Unraid requires a paid license (Lifetime tier is a \$129 one-time cost); TrueNAS SCALE is free. (The original tiebreaker — Unraid's more mature GPU-passthrough workflow — no longer applies now that the VMix VM is gone: nothing on this box needs GPU passthrough. BirdDog Central is NDI routing/control, not video encode, and runs as a plain VM. The choice stands on the container/UX grounds alone.)

**Tailscale note:** Cloud Gateway (UniFi-OS-based) generally can't run Tailscale natively without an unofficial container hack, so the mothership subnet-router role runs as a container on the Unraid box instead of the Cloud Gateway itself. See docs/open-questions.md .

## VMix nodes (x2)

Each has its own **GL-iNet A-1300** router (same model as the theatre routers) and its own Tailscale connection — a separate node on the tailnet, not part of the theatre it physically sits next to, just uplinked near it:

| Node        | Uplinks near | Subnet       | Contents                             |
|-------------|--------------|--------------|--------------------------------------|
| VMix Node 1 | Theatre 1    | 192.168.20.x | 4× BirdDog P400 cameras, 2× VMix PCs |
| VMix Node 2 | Theatre 4    | 192.168.21.x | 4× BirdDog P400 cameras, 2× VMix PCs |

That makes **14 GL-iNet A-1300s total** across the network — 12 theatre routers plus these 2. Both LAN ports are bridged together on the VMix node routers (unlike the theatre routers, there's no single traffic-sensitive device here that needs its own dedicated port the way the ATEM does).

Each VMix PC's own recording is pulled to the mothership the same way the ATEMs' recordings are — near-real-time, over this node's own dedicated uplink (not the adjacent theatre's) — see `docs/vmix-record-ingest.md` .

## Edit suite — 192.168.22.x

Post-production subsystem at the mothership: 2× MacBook Pro editing workstations, a Mac mini running the DaVinci Resolve Project Server and Remote Render, and a dedicated fast-storage NAS, on their own 10GbE LAN — **not** on the Tailscale mesh (same room as the mothership, no distance for Tailscale to bridge) and not sharing the theatre-facing network's bandwidth budget. Routed to the mothership LAN only so the ingest containers can dual-write footage to the edit-suite NAS in the same pull that feeds Nextcloud — there is no separate recording-pool-to-NAS sync step. Full design, including the storage and project-server decisions and the DIT physical-media ingest workflow, in `docs/live-editing.md` .

See `diagrams/topology.svg` for the full visual layout — every theatre, every device, every real IP — and `docs/ip-address-map.md` for the same inventory as a text table. Per-device configs: all 14 routers in `config/gl-inet/` , the Cloud Gateway in `config/unifi/` , and Tailscale in `config/tailscale-up-all-devices.sh` .